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(1) Research Program/Project/Study Title: Intensification and Optimization of Job's tears "Adlai" (Coix lacrima-jobi) Production in the Province of Capiz.	
Proponent Name:	Francisco T. Lutao, Jr. & Lailanie Abalde
Academic Rank:	
Co-Authors (with academic rank):	
Campus/Satellite College/Address:	Burias, Mambusao, Capiz
Telephone/Cell Phone/Fax Number:	
E-mail Address:	

Program or Project has component (if no proceed to 2)? ☒ Yes ☐ No

Component Title 1:	Adlai Industry in Capiz: A Retrospection
Component Leader:	Dr. Rosine O. Labado and Florence Gay B. Murguia
Academic Rank:	Assistant Professor 3

Component Title 2:	Government Policies and Interventions in Adlai Industry
Component Leader:	Dr. Aldin D. Lorenzo
Academic Rank:	Associate Professor 4

Component Title 3:	Growth and Yield Performance of three Adlai Varieties as Influenced by Integrated Nutrient Management in Burias, Mambusao, Capiz.
Component Leader:	Lailanie Abalde & Francisco T. Lutao Jr.
Academic Rank:	

Component Title 4:	Influence of Integrated Nutrient Management to the Performance of Adlai Varieties under Hilly Area in Burias, Mambusao, Capiz.
Component Leader :	Francisco T. Lutao Jr. and Lailanie Abalde
Academic Rank:	

Component Title 5:	Influence of Organic Fertilizer on the Yield and Nutrient Uptake of Adlai Intercropped with Corn.
Component Leader:	Lailanie Abalde, Francisco T. Lutao Jr. and Dr. Ryan T. Sarimong
Academic Rank:	

Component Title 6:	Impact of Post Harvest Operations on Adlay Grain Quality
Component Leader:	Francisco T. Lutao Jr., Lailanie Abalde and Eddie P. Gacasan
Academic Rank:	

Component Title 7:	Training and Techno transfer on Adlai Production
Component Leader:	Lailanie Abalde and Francisco T. Lutao
Academic Rank:	

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Component Title 9:	Impact Assessment
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Mambusao
		Province:	Capiz
		Region:	VI
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity ADLAI		(8) Discipline/Field of Specialization	

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(9) Rationale

The Philippines is considered the most food insecure in East and Southeast Asia (Global Food Security Index Q2 2022). Despite sustained economic growth, the country suffers from widespread food shortage, starvation and malnutrition. Low production in rice, corn and other agricultural products is a constant problem encountered by the country due to limited land resources, typhoons, soil fertility, land degradation and climate change.

The insufficient supply of rice and corn is not enough to feed the mouth of hungry and growing population of Filipinos, alternative staples must be considered. Adlai is the least known and taken for granted among other staple food. Adlai or katigbi (*Coix lacryma-jobi* L.), also called "Job's tears". It is a tall grain bearing tropical plant from the family Gramineae or Poaceae. In the country, four varieties are being cultivated: Gulian (white), Tapol (purple), Ginampay (brown), and Pulot (white-glutinous) (Adlai, n.d.). However, adlai is still relatively unknown to many Filipinos.

Nutritional value of adlai was reported to be comparable, or even better, than rice. It is at least 50 percent starch, 14 percent protein, and only 6 percent fat (Santos, 2012). The characteristics that make Adlai a good alternative to rice and corn showed that it provides almost the same amount of energy as milled rice (Juliano, 2003) at 356 kcal 100 g-1. Adlai is preferred by farmers from foreign countries because it tolerates in acidic soil and waterlogged area, resists pests and diseases and most of all produce a high yield in the highlands (dela Cruz, 2011).

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

The project aims to Intensify the Production of adlai In the province of Capiz and to Provide economic opportunity to the farmers of Capiz living with problematic soil condition such as Acidic soils, Marginal Lands, and to utilize the idle lands by planting Adlai Crops as a substitute for staple food such as Rice and Corn.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Component 1. Adlai Industry in Capiz: A Retrospection

Statement of the Problem

1. What are the socio - demographic characteristics of adlai industry participants such as farmers, processors, traders, and consumers in the province of Capiz?
2. What are the different practices of adlai farmers in terms of variety, planting practices, fertilizer used, pest control practices, and post-harvest facilities?
3. What are the marketing practices as to product/s or by-products produced, pricing, place, and promotion strategies utilized by key participants in the adlai industry in the province of Capiz?
4. What are the primary and support activities involved in the adlai industry in Capiz?
5. What are the different problems met by the industry participants in the sorghum industry in Capiz?

Component 2. Government Policies and Interventions in Adlai Industry

Statement of the Problem

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1. What are the policies of the government in adlai industry?
2. What are the agencies involve in the policy implementation?
3. What are the government interventions to help the adlai industry in Capiz?
4. What are the different agencies involve in the implementation of government interventions?
5. What are the problems met by the different agencies in the implementation of policies and government interventions?

Component 3. Growth and Yield Performance of three Adlai Varieties as influenced by Integrated Nutrient Management in Burias, Mambusao, Capiz.

Objectives:

1. To determine the best adlai variety suitable in Burias, Mambusao, Capiz.
2. To evaluate the growth and yield of Adlai varieties as affected by integrated nutrient Management.
3. To determine the best integrated nutrient management for adlai production
4. To assess the economic benefits of the Adlai varieties as affected by Inorganic Nutrient Management in Burias, Mambusao, Capiz

Component 4. Influence of Integrated Nutrient Management to the Performance of Adlai Varieties under Hilly Area in Burias, Mambusao, Capiz.

Objectives:

1. To evaluate the growth and yield performance of Adlai varieties as influenced by Integrated Nutrient Management under hilly area.
2. To determine the best adlai varieties for hilly areas in Burias, Mambusao, Capiz.
3. To assess the profitability of adlai varieties planted in hilly area as influenced by integrated nutrient management

Component 5. Influence of Organic Fertilizer on the Yield and Nutrient Uptake of Adlai Intercropped with Corn.

Objectives:

1. To evaluate the agronomic response of adlai as influenced by organic fertizer intercropped with corn.
2. To determine the profitability of adlai as influenced by organic fertizer intercropped with corn.

Component 6. Impact of Post Harvest Operations on Adlay Grain Quality

Objectives:

1. to identify the best post harvest practices for adlai production.

Component 7. Training and Techno transfer on Adlai Production

Objectives:

1. To Transfer the technology developed to the farmers
2. To intensify the information dissemination on adlai production

Component 8. Impact Assessment on Adlai Production in Burias, Mambusao, Capiz

Objectives:

1. To assess the Effectivity of the techno-tranfer on adlai Production
2. To evaluate the socio- economic Change of the clientele.

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(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Climate Change Rice and Adlai Production

Rice is the principal [staple](#) crop in Asia and any disturbance on rice production systems through climate change would seriously impair food security in the area. The country's palay production in the last quarter of 2018 reached to 7.16 million metric tons, lower than the production in 2017 with 7.32 million metrics tons, in as much as the yield per hectare dropped from 3.93 metric tons in 2017 to 3.87 metric tons in 2018 (Philippine Statistics Authority [PSA] 2019). As reported by Shahid (2010), climate change causes tremendous damage and negative impacts to rice production sector such as increase in temperature, increase in frequency, intensity, and duration of extreme climate events such as droughts, floods and tropical storms, soil degradation, and sea level rise resulting in loss of agricultural land. Rising sea level may amplify soil salinity, displace areas for crop production, and reduce rice production in a sizable portion of the highly productive rice land due to salt water intrusion. Crop-pest interactions may shift as the timing of development stages in both hosts and pests are altered. The possible increase in pest infestation may result to greater utilization of chemical pesticides in controlling them. Climate change may also affect weed ecology, that is, the evolution of weed species over time. Further, climate change affects irrigation water demand of rice via changes in rice physiology and phenology, soil water balances, evapotranspiration and effective precipitation, and estimation of the agricultural water demand in the changing environment.

According to Wassmann et al (2009), possible adaptation options for heat stress are derived from regions where rice crop is already exposed to very high temperature. Significant

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improvement of rice production systems that would reflect higher resilience to flooding and salinity is crucial for maintaining or even increasing yield levels.

With the current situation of climate change and its global effect in crop production that also has tantamount effect to the country and its demand for food in response to its growing population. It has to find a climate resilient and a possible substitute for rice for the people of the country not to suffer a famine. alternative source of food to its better-known cousin crops such as rice and corn. In realtion, Adlai is also known for its resilience against extreme conditions brought about by climate change.

Job's tear (*Coix Lacryma-jobi* L.), also known as Adlay/Adlai in the Philippines, is a tall grass, a grain-bearing tropical plant of the family Poaceae. Depending on the variety, the seeds are yellow, purple or brown and are often tearshaped; hence, the name "Job's tears". It has been widely studied across the globe and predominantly contains carbohydrate (65%), protein (15%), moisture (11%), fat (6%) with trace minerals and vitamins such as calcium, phosphorus, niacin, thiamine, and riboflavin; and food energy (360-380 kcal/100 g) (Mulyono et al., 2019).

Adlai is twice as rich in protein as rice, and is known among indigenous communities throughout Asia as a source of flour, coffee, tea, wine, beer and vinegar, among other products. It has anti-tumor and other medicinal properties which can help mitigate the symptoms of allergies, and diabetes. (PNRI 2017).

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(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Component 1. **Adlai Industry in Capiz: A Retrospection**

Component 2. **Government Policies and Interventions in Adlai Industry**

Component 3. **Growth and Yield Performance of three Adlai Varieties as influenced by Integrated Nutrient Management in Burias, Mambusao, Capiz.**

a. Variables to be measured and evaluated

Agronomic Characteristics

Plant Height
Number of Leaves
Number of Tillers
Percentage of Productive Tillers
Filled and Unfilled Spikelet per panicle
Days to harvest

Physiological Characteristics

LAI

Yield Characteristics

Yield / Hectare

Soil Analysis (NPK, pH)

b. Treatments to be used and Layout

V1 – Gulian
V2 – Ginampay/Kinampay
V3 – Pulot/Tapol
T0 – Control
T1 – Recommended Rate
T2 – Vermicompost
T3 – 50% RR + 50% Vermicompost
T4 – BOKASHI
T5 – 50% RR + 50% BOKASHI

c. Experimental Procedures and Design

The experiment will be laid out in a randomized complete block design with three replications. With a planting distance of 90cm x 60cm and two seeds per hill.

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d. Statistical Analysis

After gathering data, the means will be computed and will be analyzed using analysis of variance (ANOVA) will be done using Statistical Analysis System (SAS Version 6.12). Comparison of means was done using Honestly Significant Difference (HSD) or Tukey's test.

e. Evaluation method and observations to be made, strategies for implementation

Component 4. **Influence of Integrated Nutrient Management to the Performance of Adlai Varieties under Hilly Area in Burias, Mambusao, Capiz.**

a. Variables to be measured and evaluated

Agronomic Characteristics

Plant Height
Number of Leaves
Number of Tillers
Percentage of Productive Tillers
Filled and Unfilled Spikelet per panicle
Days to harvest

Physiological Characteristics

LAI

Yield Characteristics

Yield / Hectare

Soil Analysis (NPK, pH)

b. Treatments to be used and Layout

V1 – Gulian
V2 – Ginampay/Kinampay
V3 – Pulot/Tapol
T0 – Control
T1 – Recommended Rate
T2 – Vermicompost
T3 – 50% RR + 50% Vermicompost
T4 – BOKASHI
T5 – 50% RR + 50% BOKASHI

c. Experimental Procedures and Design

The experiment will be laid out in a randomized complete block design with three replications. The planting distance of 90cm x 60cm and two seeds per hill.

d. Statistical Analysis

After gathering data, the means will be computed and will be analyzed using analysis of variance (ANOVA) will be done using Statistical Analysis System (SAS Version 6.12). Comparison of means was done using Honestly Significant Difference (HSD) or Tukey's test.

e. evaluation method and observations to be made, strategies for implementation

Component 5. **Influence of Organic Fertilizer on the Yield and Nutrient Uptake of Adlai Intercropped with Corn.**

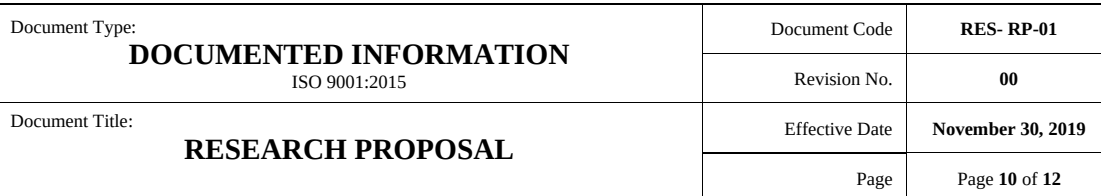
a. Variables to be measured and evaluated

Agronomic Characteristics

Plant Height

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Number of Leaves Number of Tillers Percentage of Productive Tillers Filled and Unfilled Spikelet per panicle Days to harvest Physiological Characteristics LAI Yield Characteristics Yield / Hectare Soil Analysis (NPK, pH) Plant Analysis Nitrogen Potassium Phosphorous b. Treatments to be used and Layout V1 – Gulian V2 – Ginampay/Kinampay V3 – Pulot/Tapol M1 – Monocrop Adlai M2 – Adlai Intercrop with corn T0 – Control T1 – Vermicompost T2 – Bokashi T3 – Commercial Organic Fertilizer c. Experimental Procedures and Design The experiment will be laid out in a randomized complete block design with three replications. The planting distance of 90cm x 60cm and two seeds per hill. d. Statistical Analysis After gathering data, the means will be computed and will be analyzed using analysis of variance (ANOVA) will be done using Statistical Analysis System (SAS Version 6.12). Comparison of means was done using Honestly Significant Difference (HSD) or Tukey's test. e. evaluation method and observations to be made, strategies for implementation Component 6. Impact of Post Harvest Operations on Adlay Grain Quality a. Variables to be measured and evaluated Physical Characteristics: 1. Visual Quality Rating 2. Yellowing Index 3. Firmness 4. Softening Index 5. Color 6. Shelf-life Chemical Characteristics 1. Amylopectin 2. Amylose b. Treatments to be used and Layout V1 – Gulian V2 – Ginampay/Kinampay V3 – Pulot/Tapol
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Component 9. Impact Assessment

[illegible]

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1. Field work/observation												
2. Analysis of data gathered from observation												
3. Lay-outing, Editing and Printing of 100 copies of												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Training and Techno transfer on Adlai Production

- **Adlai Production Management**
- **Marketing of Adlai Products**

(16) Targets Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

Farmers of marginalized areas and idle lands in Capiz.

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Francisco T. Luta	Project Leader	Crop Science
Lailanie Abalde	Project Member	Crop Science

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

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(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	3	1, 404, 000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		800, 000.00
- Meal allowance during data gathering		300,000.00
- Professional fees (laboratory analysis and other related fees)		2,200,000.00
- Miscellaneous (for incidental expenses)		1, 000, 000.00
- Printing		600,000.00
C. CAPITAL OUTLAY		2, 000,000.00
Total		8, 304,000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Dela Cruz, RT (2011). Champion Crop of the Subanen Tribe. BAR Chronicle. Vol 12 No. 7

Juliano BO (2003) Rice Chemistry and Quality. Manila. Philippine Rice Research Institute, Los Baños, Laguna.

Rice and Corn Situation and Outlook, January 2019 Round. psa.gov.ph

Mulyono, E., Kusuma, A., Dewandari, K. T., & Darniadi, S. (2019, September). Preliminary Study of Hanjeli (*Coix lacryma-jobi* L) Flour for Food Uses. In IOP Conference Series: Earth and Environmental Science (Vol. 309, No. 1, p. 012057). IOP Publishing

Mutant Adlai as Alternative Staple Food Crop for Filipinos. DOST-PNRI 2017.

Shahid, S (2010) Impact of Climate Change on Irrigation Water Demand of Dry Season Boro Rice in Northwest Bangladesh. Climatic Change, 105, 433-453.

Wassmann, R.; Jagadish, S.V.K.; Heuer, S.; Ismail, A.; Redona, E.; Serraj, R.; Singh, R.K.; Howell, G.; Pathak, H.; Sumfleth, K. Climate change affecting rice production: The physiological and agronomic basis for possible adaptation strategies. *Adv. Agron.* 2009, **101**, 59–122.

(20) Capsule Curriculum Vitae (One-page brief C.V. only)